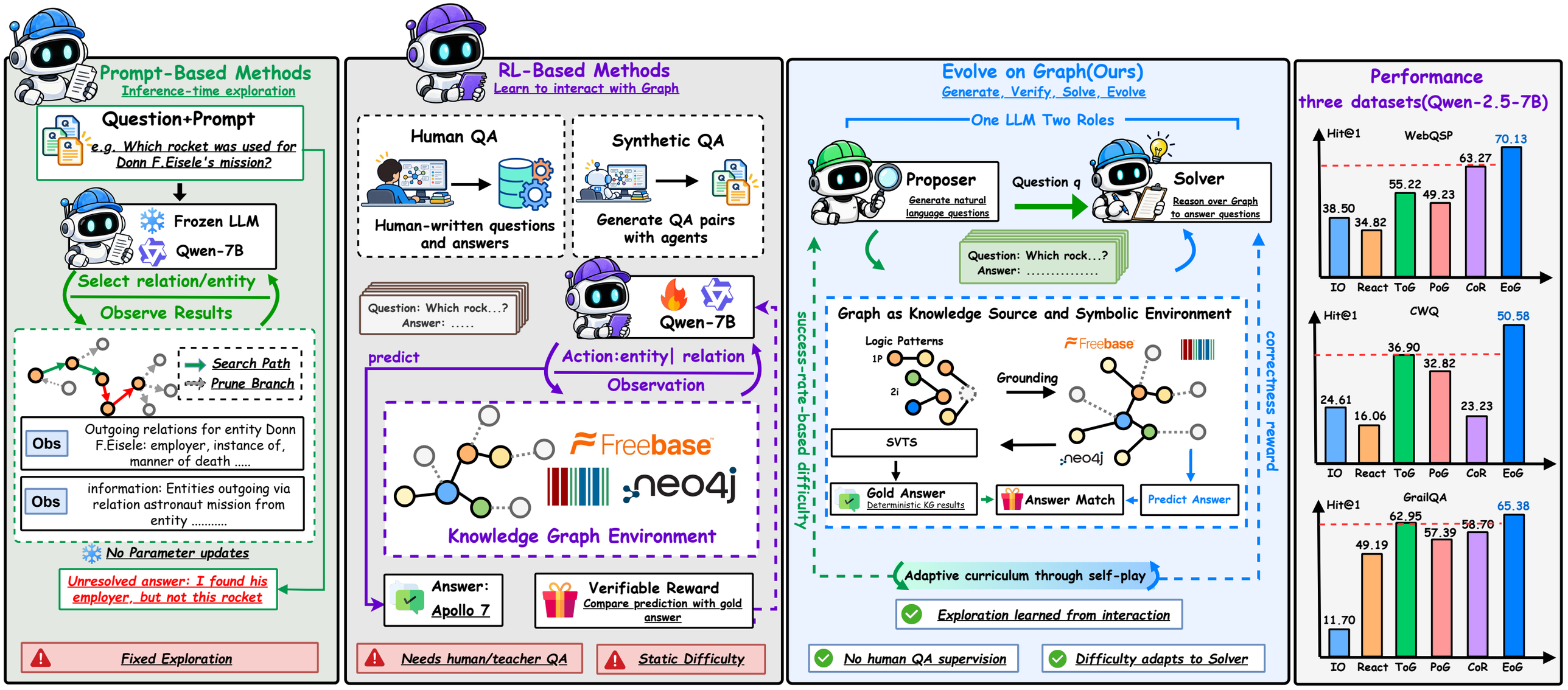

Prompt-Based Methods
Inference-time exploration
Question+Prompt
e.g. Which rocket was used for Donn F.Eisele's mission?
Frozen LLM
Qwen-7B
Select relation/entity
Observe Results
Search Path
Prune Branch
Obs
Outgoing relations for entity Donn F.Eisele: employer, instance of, manner of death .....
Obs
information: Entities outgoing via relation astronaut mission from entity .....
No Parameter updates
Unresolved answer: I found his employer, but not this rocket
Fixed Exploration
RL-Based Methods
Learn to interact with Graph
Human QA
Human-written questions and answers
Synthetic QA
Generate QA pairs with agents
Question: Which rock...?
Answer: .....
Qwen-7B
predict
Action:entity| relation
Observation
Freebase
neo4j
Knowledge Graph Environment
Answer:
Apollo 7
Verifiable Reward
Compare prediction with gold answer
Needs human/teacher QA
Static Difficulty
Evolve on Graph(Ours)
Generate, Verify, Solve, Evolve
One LLM Two Roles
Proposer
Generate natural language questions
Question q
Solver
Reason over Graph to answer questions
Question: Which rock...?
Answer: .....
Graph as Knowledge Source and Symbolic Environment
Logic Patterns
1P
2i
SVTS
Grounding
Freebase
neo4j
Gold Answer
Deterministic KG results
Answer Match
Predict Answer
success-rate-based difficulty
correctness reward
Adaptive curriculum through self-play
Exploration learned from interaction
No human QA supervision
Difficulty adapts to Solver
Performance
three datasets(Qwen-2.5-7B)
Hit@1
WebQSP
38.50
34.82
55.22
49.23
63.27
70.13
IO
React
ToG
PoG
CoR
EoG
Hit@1
CWQ
24.61
16.06
36.90
32.82
23.23
50.58
IO
React
ToG
PoG
CoR
EoG
Hit@1
GrailQA
11.70
49.19
62.95
57.39
58.70
65.38
IO
React
ToG
PoG
CoR
EoG